TOPIC 1: INTRODUCTION TO OPERATING SYSTEMS

1.1 What is an Operating System?

An Operating System (OS) is system software that manages computer hardware and software resources and provides an environment in which application programs can run.

The operating system acts as an intermediary between the user and computer hardware.

Simple representation

USER

↓

APPLICATION SOFTWARE

↓

OPERATING SYSTEM

↓

COMPUTER HARDWARE

WHATSAPP:0720349953

Examples of Operating Systems

- Microsoft Windows
- Linux
- macOS
- Android

- iOS
- ChromeOS
- Unix

1.2 Functions of an Operating System

1. Process Management

The OS manages programs that are currently running.

It:

- Creates processes
- Terminates processes
- Allocates CPU time
- Schedules processes
- Handles process synchronization

2. Memory Management

The OS manages the computer's main memory (RAM).

It determines:

- Which programs are loaded into memory

- How much memory each process receives
- When memory is released

3. File Management

The OS manages files and directories.

It allows users to:

- Create files
- Delete files
- Rename files
- Copy files
- Move files
- Organize files into directories

4. Device Management

The OS manages input and output devices such as:

- Keyboard
- Mouse
- Printer
- Monitor
- Hard disk
- USB devices

5. Security Management

The OS protects computer resources from unauthorized access.

Examples include:

- Passwords
- User accounts
- Permissions
- Encryption
- Access control

6. User Interface

The OS provides an interface through which users interact with the computer.

Two major interfaces are:

CLI — Command Line Interface

C:\Users\Student>

GUI — Graphical User Interface

Uses:

- Windows
- Icons
- Menus
- Pointers
- Buttons

1.3 Characteristics of an Operating System

A good operating system should provide:

- Reliability
- Security
- Efficiency
- Convenience

- Resource management
- Error handling
- User friendliness
- Multitasking

1.4 Types of Operating Systems

Batch Operating System

Jobs are collected and processed in batches without direct interaction from the user.

Time-Sharing Operating System

CPU time is shared among multiple users or processes.

Real-Time Operating System

Responds to events within a specified time.

Examples:

- Industrial control systems
- Medical equipment
- Automotive systems

Distributed Operating System

Manages multiple connected computers and makes them appear as one system.

Network Operating System

Provides services for computers connected to a network.

Mobile Operating System

Designed for smartphones and tablets.

Examples:

- Android

- iOS

Revision Questions

1. Define an operating system.
2. State five functions of an operating system.
3. Give four examples of operating systems.
4. Differentiate between CLI and GUI.
5. Explain four types of operating systems.

TOPIC 2: OPERATING SYSTEM STRUCTURES

2.1 Components of an Operating System

An operating system consists of several major components.

Main components

1. Kernel
2. Shell
3. File system
4. Device management
5. Process management
6. Memory management
7. Security management
8. User interface

2.2 Kernel

The kernel is the central component of an operating system.

It communicates directly with hardware and manages system resources.

Major functions of the kernel

- Process management
- Memory management
- Device management
- File management
- Security
- Hardware communication

2.3 Shell

The shell provides an interface between the user and the operating system.

Examples:

- Windows Command Prompt
- PowerShell
- Bash
- Zsh

Example:

mkdir students

The command tells the operating system to create a directory called students.

2.4 System Calls

A system call is a mechanism through which a program requests a service from the operating system.

Examples of services include:

- Creating a process

- Opening a file
- Reading a file
- Writing to a file
- Allocating memory

Common categories of system calls

Category Purpose

Process control Create/terminate processes

File management Open/read/write files

Device management Control devices

Information maintenance Obtain system information

Communication Allow processes to communicate

Protection Manage permissions

2.5 Operating System Structures

Common OS structures include:

Monolithic structure

Most OS services operate inside the kernel.

Layered structure

The OS is divided into layers.

User Applications

↓

User Interface

↓

File Management

↓

Memory Management

↓

Hardware

Microkernel

Only essential services operate inside the kernel while other services run outside it.

Modular structure

The OS uses separate modules that can be loaded when needed.

Revision Questions

1. What is a kernel?
2. State four functions of the kernel.
3. Define a system call.
4. Give four examples of system-call categories.
5. Differentiate between a shell and a kernel.

TOPIC 3: PROCESS MANAGEMENT

3.1 What is a Process?

A process is a program that is currently being executed.

For example:

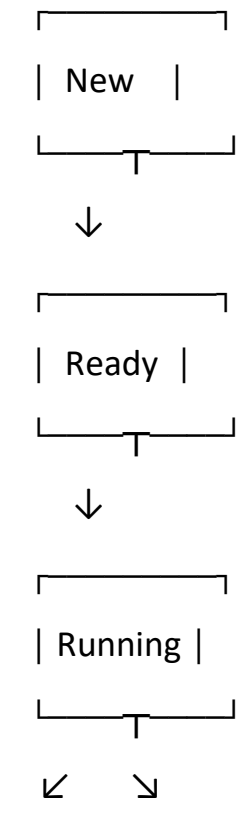
A program stored on disk is not necessarily a process. When it is loaded into memory and begins executing, it becomes a process.

3.2 Program vs Process

Program	Process
Passive	Active
Stored on storage	Executing in memory
Contains instructions	Instructions are being executed
Does not have execution state	Has an execution state

3.3 Process States

A process commonly passes through several states.



New

The process is being created.

Ready

The process is waiting for CPU allocation.

Running

The CPU is executing the process.

Waiting/Blocked

The process is waiting for an event or resource.

Terminated

The process has finished execution.

3.4 Process Control Block

A Process Control Block (PCB) is a data structure used by the operating system to store information about a process.

It may contain:

- Process ID
- Process state
- Program counter
- CPU registers
- CPU scheduling information
- Memory-management information
- Accounting information
- I/O status information

3.5 Context Switching

A context switch occurs when the CPU changes from executing one process to another.

The OS saves the state of the current process and loads the state of another process.

Example:

Process A

↓

Save A's context

↓

Load B's context

↓

Process B

3.6 Threads

A thread is the smallest unit of CPU execution within a process.

A process can contain multiple threads.

Advantages of threads

- Faster execution
- Better responsiveness
- Resource sharing
- Efficient multitasking

Revision Questions

1. Define a process.
2. Differentiate between a program and a process.
3. State five process states.
4. What is a PCB?

5. Explain context switching.

6. What is a thread?

TOPIC 4: CPU SCHEDULING

4.1 What is CPU Scheduling?

CPU scheduling is the process of selecting which ready process should be given CPU time.

The CPU scheduler selects processes from the ready queue.

4.2 Scheduling Objectives

CPU scheduling aims to:

- Maximize CPU utilization
- Maximize throughput
- Minimize waiting time
- Minimize turnaround time
- Minimize response time
- Provide fairness

4.3 First Come First Served

FCFS executes processes in the order in which they arrive.

Example:

P1 → P2 → P3

Advantages

- Simple
- Easy to implement

- Fair based on arrival order

Disadvantages

- Can produce long waiting times
- Convoy effect can occur

4.4 Shortest Job First

SJF selects the process with the shortest CPU burst.

Example:

Process Burst Time

P1 8

P2 3

P3 5

Order:

P2 → P3 → P1

4.5 Priority Scheduling

Each process receives a priority.

The CPU executes the process with the highest priority according to the scheduling convention being used.

4.6 Round Robin

Round Robin assigns each process a fixed time interval called a time quantum.

Example:

P1 → P2 → P3 → P1 → P2 → P3

It is commonly associated with time-sharing systems.

4.7 Scheduling Terms

Waiting Time

Time a process spends waiting in the ready queue.

Turnaround Time

Turnaround Time =

Completion Time - Arrival Time

Response Time

Time from submission of a request until the system produces the first response.

Revision Questions

1. Define CPU scheduling.
2. Explain FCFS.
3. Explain SJF.
4. What is Round Robin scheduling?
5. State four CPU scheduling objectives.
6. Define waiting time.

TOPIC 5: MEMORY MANAGEMENT

5.1 What is Memory Management?

Memory management is the function of the operating system responsible for managing the computer's main memory.

The OS:

- Allocates memory
- Deallocates memory

- Tracks memory usage
- Protects memory
- Manages virtual memory

5.2 Main Memory

Main memory is commonly referred to as RAM.

Programs must generally be loaded into RAM before the CPU can execute them.

5.3 Memory Allocation

Memory allocation involves assigning memory space to processes.

Contiguous allocation

A process is placed in one continuous block of memory.

Non-contiguous allocation

A process can occupy multiple memory locations.

5.4 Paging

Paging divides:

- Physical memory into frames
- Logical memory into pages

A process's pages can be placed into available frames.

Logical Memory

Page 0

Page 1

Page 2

Page 3

↓

Physical Memory

Frame 5 → Page 0

Frame 2 → Page 1

Frame 8 → Page 2

Frame 1 → Page 3

Advantage

Paging reduces external fragmentation.

Disadvantage

Paging can cause internal fragmentation and requires page-table management.

5.5 Segmentation

Segmentation divides a program into logical sections such as:

- Code
- Data
- Stack
- Procedures

Each segment has a logical meaning.

5.6 Virtual Memory

Virtual memory allows a system to use storage space as an extension of RAM.

It allows programs larger than available physical memory to execute by moving portions between RAM and storage.

5.7 Page Fault

A page fault occurs when a process references a page that is not currently in physical memory.

The OS must retrieve the required page from secondary storage.

5.8 Page Replacement

When memory is full, the OS may need to replace an existing page.

Common algorithms include:

- FIFO
- Optimal
- LRU

Revision Questions

1. Define memory management.
2. What is paging?
3. Differentiate between a page and a frame.
4. Explain virtual memory.
5. What is a page fault?
6. State three page replacement algorithms.

TOPIC 6: FILE MANAGEMENT

6.1 What is a File?

A file is a named collection of related data stored on a storage device.

Examples:

student.pdf

marks.xlsx

photo.jpg

program.cpp

notes.docx

6.2 File Attributes

File attributes may include:

- Name
- Type
- Size
- Location
- Protection
- Creation date
- Modification date
- Owner

6.3 File Operations

An operating system can perform:

- Create
- Open
- Read

- Write
- Rename
- Copy
- Move
- Delete
- Close

6.4 Directory

A directory is a structure used to organize files.

Example:

TVET-HUB



6.5 File Allocation Methods

Contiguous Allocation

Files occupy consecutive blocks.

Linked Allocation

File blocks can be located anywhere and linked together.

Indexed Allocation

An index block contains pointers to file blocks.

6.6 File Permissions

Operating systems can control who can:

- Read
- Write
- Execute

Example in Linux:

r = read

w = write

x = execute

Revision Questions

1. Define a file.
2. State five file attributes.
3. List six file operations.
4. What is a directory?
5. Explain three file allocation methods.
6. What do r, w, and x represent?

TOPIC 7: DEADLOCKS

7.1 What is Deadlock?

A deadlock is a situation where two or more processes are permanently waiting for resources held by each other.

Example:

Process A holds Resource 1

↓

waits for Resource 2

Process B holds Resource 2

↓

waits for Resource 1

Neither process can continue.

7.2 Conditions for Deadlock

Four conditions must generally exist simultaneously.

1. Mutual Exclusion

A resource can be used by only one process at a time.

2. Hold and Wait

A process holds at least one resource while waiting for another.

3. No Preemption

A resource cannot be forcibly taken from a process.

4. Circular Wait

Processes form a circular chain of waiting.

P1 → P2 → P3 → P1

7.3 Deadlock Prevention

Deadlock prevention attempts to ensure that at least one of the necessary conditions cannot occur.

Methods include:

- Preventing hold and wait
- Allowing resource preemption
- Imposing resource ordering

7.4 Deadlock Avoidance

Deadlock avoidance checks whether allocating a resource will leave the system in a safe state.

One important method is the:

Banker's Algorithm

The algorithm evaluates whether granting a request could result in an unsafe state.

7.5 Deadlock Detection

The OS can periodically check whether deadlock exists.

If detected, the OS can recover using methods such as:

- Terminating processes
- Resource preemption
- Rolling back processes

Revision Questions

1. Define deadlock.

2. State the four conditions necessary for deadlock.
3. Explain deadlock prevention.
4. What is deadlock avoidance?
5. What is the Banker's Algorithm?
6. State three methods of deadlock recovery.

TOPIC 8: INPUT/OUTPUT MANAGEMENT

8.1 What is I/O Management?

Input/Output management involves controlling communication between the computer and peripheral devices.

Input devices

- Keyboard
- Mouse
- Scanner
- Microphone
- Camera

Output devices

- Monitor
- Printer
- Speakers
- Projector

8.2 Device Controller

A device controller is hardware that manages communication between a peripheral device and the CPU/system.

8.3 Device Driver

A device driver is software that allows the operating system to communicate with a specific hardware device.

Examples:

- Printer driver
- Graphics driver
- Audio driver
- Network driver

8.4 Interrupts

An interrupt is a signal that causes the CPU to temporarily stop its current execution and handle an event.

Example:

When a keyboard key is pressed, the keyboard can generate an interrupt.

8.5 Buffering

Buffering temporarily stores data while it is being transferred between devices or processes.

Example:

A printer can receive data into a buffer while the computer continues performing other work.

8.6 Spooling

Spooling stores jobs temporarily so that a slower device can process them sequentially.

A common example is print spooling.

Computer

↓

Print Queue

↓

Printer

Revision Questions

1. Define I/O management.
2. Give four input devices.
3. Give four output devices.
4. What is a device driver?
5. Explain an interrupt.
6. Differentiate between buffering and spooling.

TOPIC 9: INTER-PROCESS COMMUNICATION

9.1 What is IPC?

Inter-Process Communication (IPC) refers to mechanisms that allow processes to communicate and exchange data.

9.2 Why is IPC Important?

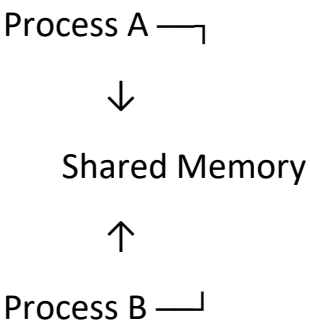
IPC allows processes to:

- Exchange information
- Coordinate activities
- Share resources

- Synchronize execution

9.3 Shared Memory

Processes communicate by accessing a shared memory area.



Advantage

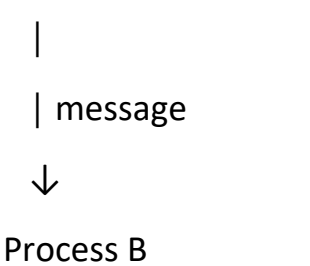
Fast communication.

Disadvantage

Requires synchronization to avoid data conflicts.

9.4 Message Passing

Processes communicate by sending and receiving messages.



It can be implemented using:

- Send
- Receive

9.5 Pipes

A pipe provides a communication channel between processes.

For example, in Linux:

ls | grep ".txt"

The output of ls is passed to grep.

9.6 Synchronization

Synchronization ensures that processes access shared resources in a controlled manner.

It helps prevent problems such as:

- Race conditions
- Data inconsistency

9.7 Peterson's Solution

Peterson's solution is a software-based algorithm for mutual exclusion between two processes.

It uses shared variables to control which process can enter a critical section.

9.8 Critical Section

A critical section is a part of a program where shared resources are accessed.

Only one process should normally execute a critical section at a time when mutual exclusion is required.

Revision Questions

1. Define IPC.
2. State three methods of IPC.
3. Explain shared memory.
4. Explain message passing.
5. What is synchronization?
6. Define a critical section.
7. What problem does Peterson's solution address?

TOPIC 10: SECURITY AND PROTECTION

10.1 Operating System Security

OS security involves protecting the operating system and its resources from unauthorized access, damage and misuse.

10.2 Authentication

Authentication verifies the identity of a user.

Examples:

- Password
- PIN
- Fingerprint
- Facial recognition
- Security token

10.3 Authorization

Authorization determines what an authenticated user is allowed to access.

Example:

Student → View notes

Teacher → Upload notes

Administrator → Manage users

10.4 Authentication vs Authorization

Authentication	Authorization
Verifies identity	Determines permissions
"Who are you?"	"What can you access?"
Happens before authorization	Usually follows authentication

10.5 Access Control

Access control determines which users or processes can access system resources.

Examples:

- File permissions
- User roles
- Access Control Lists

10.6 Malware

Malware is malicious software designed to damage, disrupt, spy on or gain unauthorized access to systems.

Examples:

- Virus
- Worm
- Trojan horse
- Ransomware

- Spyware

10.7 Protection Measures

Users can improve OS security by:

- Using strong passwords
- Installing updates
- Using reputable security software
- Backing up important data
- Avoiding suspicious downloads
- Using multi-factor authentication
- Restricting user privileges

10.8 Encryption

Encryption converts readable data into an encoded form.

Plaintext

↓

Encryption

↓

Ciphertext

Decryption converts the ciphertext back into readable information.

Revision Questions

1. Define authentication.
2. Define authorization.

3. Differentiate authentication from authorization.
4. State five examples of malware.
5. State five ways of protecting an operating system.
6. What is encryption?

TOPIC 11: STORAGE MANAGEMENT

11.1 Secondary Storage

Secondary storage is used to store data permanently.

Examples:

- HDD
- SSD
- USB flash drive
- Memory card
- Optical disc

11.2 Hard Disk Drive

An HDD stores data magnetically on rotating platters.

Advantages

- Large storage capacity
- Relatively inexpensive

Disadvantages

- Mechanical parts
- Slower than many SSDs
- More vulnerable to physical shock

11.3 Solid-State Drive

An SSD uses flash memory and has no moving mechanical platters.

Advantages

- Fast
- Low latency
- Quiet
- More resistant to physical shock

11.4 Disk Scheduling

Disk scheduling determines the order in which disk requests are serviced.

Common algorithms include:

- FCFS
- SSTF
- SCAN
- C-SCAN
- LOOK
- C-LOOK

11.5 RAID

RAID stands for Redundant Array of Independent Disks.

It combines multiple physical disks to provide different combinations of performance, capacity and/or redundancy.

RAID 0

Uses striping.

Advantage: High performance.

Disadvantage: No redundancy.

RAID 1

Uses mirroring.

Disk 1 → Data A

Disk 2 → Data A

Provides redundancy but uses more storage capacity.

RAID 5

Uses striping with distributed parity.

It can tolerate the failure of one disk in the array.

Revision Questions

1. Define secondary storage.
2. Give five examples of secondary storage.
3. Differentiate between HDD and SSD.
4. What is disk scheduling?
5. Name four disk scheduling algorithms.
6. Define RAID.
7. Explain RAID 0 and RAID 1.

TOPIC 12: COMMON OPERATING SYSTEMS

12.1 Microsoft Windows

Windows is a graphical operating system developed by Microsoft.

Examples include:

- Windows 10
- Windows 11
- Windows Server

Features

- GUI
- File Explorer
- Multitasking
- User accounts
- Networking
- Device management

12.2 Linux

Linux is a family of operating systems built around the Linux kernel.

Popular distributions include:

- Ubuntu
- Debian
- Fedora
- Linux Mint
- Kali Linux

Features

- Open-source ecosystem
- Multiuser capabilities
- Multitasking
- Strong command-line tools

- Networking capabilities

12.3 macOS

macOS is Apple's desktop operating system for Mac computers.

Features include:

- Graphical interface
- Multitasking
- File management
- Security features
- Integration with Apple hardware and services

12.4 Android

Android is a mobile operating system widely used on smartphones, tablets and other devices.

Features include:

- Touch interface
- Mobile applications
- Notifications
- Wireless networking
- Multitasking

12.5 iOS

iOS is Apple's mobile operating system used on iPhone devices.

Features include:

- Touch interface

- App ecosystem
- Security controls
- Multitasking features
- Integration with Apple services

Comparison

OS	Common Use
Windows	PCs and laptops
Linux	Servers, PCs, embedded systems
macOS	Mac computers
Android	Smartphones/tablets
iOS	iPhones

Revision Questions

1. What is Windows?
2. Give three Linux distributions.
3. State four features of Linux.
4. What is Android?
5. Differentiate between Android and iOS.
6. Give two examples of desktop operating systems.

TOPIC 13: OPERATING SYSTEM INSTALLATION AND MAINTENANCE

13.1 OS Installation

Operating system installation involves preparing a computer and installing the required operating system.

General installation process

Check Requirements

↓

Backup Data

↓

Create Installation Media

↓

Boot from Installation Media

↓

Partition/Select Storage

↓

Install OS

↓

Install Drivers

↓

Install Updates

↓

Configure System

13.2 System Requirements

Before installation, check:

- Processor requirements
- RAM requirements
- Storage requirements

- Graphics requirements
- Firmware/boot requirements
- Network requirements where applicable

13.3 Partitioning

Partitioning divides a storage device into logical sections.

Example:

1 TB Drive

C: System

D: Documents

E: Backup

13.4 Formatting

Formatting prepares a storage partition for a file system.

Common file systems include:

- NTFS
- FAT32
- exFAT
- ext4

13.5 Device Drivers

After installation, appropriate drivers may be needed for:

- Network adapter

- Graphics
- Audio
- Printer
- Chipset
- Other hardware

13.6 OS Updates

Updates can provide:

- Security fixes
- Bug fixes
- Performance improvements
- New features
- Hardware compatibility improvements

13.7 Backup and Recovery

A backup is a copy of important data stored separately from the original.

Examples:

- External drive
- Network storage
- Cloud storage

Recovery allows a system or data to be restored after problems occur.

13.8 OS Troubleshooting

Common problems include:

Computer is slow

Possible causes:

- Too many background programs
- Insufficient memory
- Low storage space
- Malware
- Outdated software

OS fails to boot

Possible causes:

- Damaged system files
- Storage failure
- Boot configuration problems
- Hardware problems

No internet

Possible causes:

- Network adapter problem
- Driver problem
- Router problem
- Incorrect network settings

OS Maintenance Best Practices

- Keep the OS updated
- Back up important data
- Remove unnecessary software

- Monitor storage space
- Use appropriate security controls
- Keep drivers updated where necessary
- Avoid untrusted software
- Restart when required after updates

Revision Questions

1. List the major steps in OS installation.
2. What is partitioning?
3. Define formatting.
4. Give four examples of file systems.
5. Why are device drivers important?
6. State five OS maintenance practices.
7. Give three causes of OS boot failure.

TOPIC 14: REVISION QUESTIONS AND PRACTICAL ACTIVITIES

SECTION A: MULTIPLE CHOICE QUESTIONS

1. What is an operating system?

A. Application software

B. System software

C. Programming language

D. Database

Answer: B

2. Which component is the core of an operating system?

A. Browser

B. Shell

C. Kernel

D. Application

Answer: C

3. Which memory is directly managed for active processes?

A. RAM

B. DVD

C. Flash disk

D. ROM

Answer: A

4. Which scheduling algorithm uses a time quantum?

A. FCFS

B. SJF

C. Round Robin

D. Priority

Answer: C

5. Which condition is associated with deadlock?

A. Mutual exclusion

B. Compilation

C. Encryption

D. Formatting

Answer: A

6. Which is an example of IPC?

A. Shared memory

B. Formatting

C. Partitioning

D. Defragmentation

Answer: A

7. Which RAID level uses mirroring?

A. RAID 0

B. RAID 1

C. RAID 5

D. RAID 10 only

Answer: B

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8. Which of the following is a Linux distribution?

A. Ubuntu

B. Excel

C. Chrome

D. Photoshop

Answer: A

_____

9. What does authentication verify?

A. File size

B. User identity

C. CPU speed

D. Disk capacity

Answer: B

_____

10. Which technology allows storage to act as an extension of RAM?

A. Virtual memory

B. RAID

C. BIOS

D. Spooling

Answer: A

SECTION B: SHORT-ANSWER QUESTIONS

1. Define an operating system.
2. State six functions of an operating system.
3. Explain the difference between system software and application software.
4. Define a process.
5. Explain the purpose of a Process Control Block.
6. What is CPU scheduling?
7. Explain FCFS scheduling.
8. Explain Round Robin scheduling.

9. Define virtual memory.
10. What is paging?
11. Define a file.
12. What is a directory?
13. Define deadlock.
14. State four necessary conditions for deadlock.
15. Explain deadlock prevention.
16. Define IPC.
17. Explain shared memory.
18. What is an interrupt?
19. Define authentication and authorization.
20. What is RAID?
21. Differentiate between HDD and SSD.
22. What is a device driver?
23. Define a page fault.
24. What is a critical section?
25. Explain context switching.

SECTION C: STRUCTURED QUESTIONS

Question 1

a) Define an operating system.

b) State five functions of an operating system.

c) Explain three types of operating systems.

Suggested answer

a) An operating system is system software that manages computer hardware and software resources and provides services to application programs.

b)

- Process management
- Memory management
- File management
- Device management
- Security management

- Batch OS
- Real-time OS
- Time-sharing OS

Question 2

A computer is running several programs simultaneously.

a) What is a process?

b) Explain the process states.

c) What is context switching?

Question 3

a) Define deadlock.

b) State the four necessary conditions for deadlock.

c) Explain two methods of handling deadlocks.

Question 4

a) Define virtual memory.

b) Explain paging.

c) What is a page fault?

d) State three page replacement algorithms.

Question 5

a) What is IPC?

b) Explain shared memory.

c) Explain message passing.

d) State two reasons why process synchronization is important.

PRACTICAL ACTIVITIES FOR TVET STUDENTS

Practical 1: Windows File Management

Students should practice:

1. Creating a folder.
2. Creating subfolders.
3. Renaming files.
4. Copying files.
5. Moving files.
6. Deleting files.
7. Restoring deleted files from the Recycle Bin.

Practical 2: Windows Task Manager

Open Task Manager and identify:

- Running processes
- CPU usage
- Memory usage
- Disk usage
- Network usage

Students should explain what happens when a program consumes excessive CPU or memory.

Practical 3: Linux Basic Commands

Practice commands such as:

pwd

ls

cd

mkdir

touch

cp

mv

rm

cat

Example:

mkdir students

cd students

touch notes.txt

ls

Practical 4: Process Management

Students should identify running processes using the operating system's process-management tools.

They should record:

Process	CPU Usage	Memory Usage
Browser	—	—
File Manager	—	—
Text Editor	—	—

QUICK OS GLOSSARY

Term	Meaning
OS	Operating System
CPU	Central Processing Unit
RAM	Random Access Memory
ROM	Read-Only Memory
GUI	Graphical User Interface
CLI	Command Line Interface
PCB	Process Control Block
IPC	Inter-Process Communication
I/O	Input/Output
HDD	Hard Disk Drive
SSD	Solid-State Drive
RAID	Redundant Array of Independent Disks
SJF	Shortest Job First

FCFS First Come First Served

FIFO First In, First Out

LRU Least Recently Used

FINAL REVISION CHECKLIST

Before an examination, a student should be able to explain:

- Definition of an operating system
- Functions of an operating system
- Types of operating systems
- Kernel and shell
- System calls
- Processes and process states
- Process Control Block
- Context switching
- Threads
- CPU scheduling
- FCFS
- SJF
- Priority scheduling
- Round Robin
- Memory management
- Paging
- Segmentation
- Virtual memory

- Page replacement
- File management
- File allocation
- Deadlocks
- Banker's Algorithm
- I/O management
- Interrupts
- Buffering and spooling
- IPC
- Shared memory
- Message passing
- Synchronization
- Peterson's solution
- OS security
- Authentication and authorization
- Malware
- Storage management
- Disk scheduling
- RAID
- Windows, Linux, macOS, Android and iOS
- OS installation
- Partitioning and formatting
- Drivers
- OS maintenance and troubleshooting